

ASSIGNMENT 4 / OPTION A

Defend Your Thesis

An AI-assisted HTML5 Canvas game where a graduate student protects a thesis core and committee gate from bugs, deadlines, and peer reviewers.

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Thesis Defense Game

GOAL

From classroom prompt to stable web application

Assignment Requirement

- Develop a functional software application.
- Use LLMs as the primary development partner.
- Deliver a hosted web service or executable application.
- Document architecture, problem solving, and reflection.

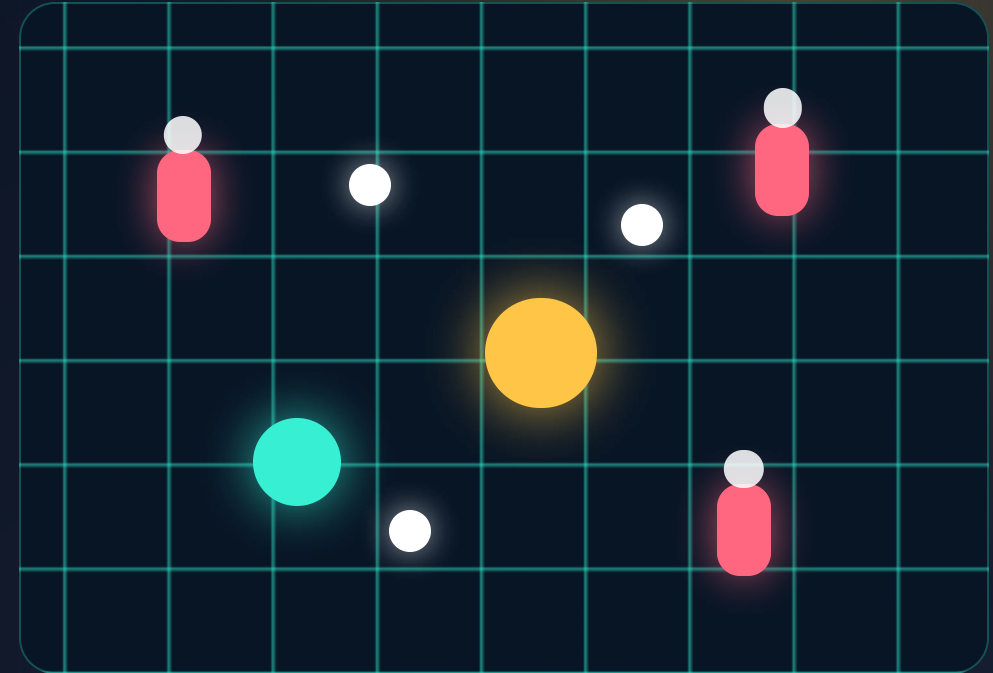
My Delivery

- **Static hosted game** on GitHub Pages.
- Playable from the Assignment 4 page.
- Report PDF and Markdown documentation included.

Protect the thesis core under academic pressure

Game Theme

- The player is a graduate student defender.
- The center objective is the thesis core and defense gate.
- Bugs attack the Code Lab, Deadlines attack the Schedule Gate, and Peer Reviewers attack the Committee Door.
- Gate archers shoot matching attackers from range.
- If an archer is pinned down, the player must rescue them with a sword.
- Citations are collectibles that increase the final score.



- Successful defense weakens threats; failed defense makes them stronger.

Short, playable, and easy to demonstrate

Key Features

- Character selection screen with seven playable roles.
- Keyboard controls with Arrow keys or WASD.
- Mouse and touch movement by clicking the canvas.
- Score, defended count, integrity, rejection gate, timer, and threat strength HUD.
- Gate archer helpers, arrow projectiles, and sword rescue moments.
- Clear pass/fail logic: survive 60 seconds, or lose by collapse/rejection.

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PLAYABLE CHARACTERS

3

THREAT TYPES

60s

DEFENSE ROUND

2

INPUT MODES

Static HTML, CSS, and JavaScript

Core Files

- **thesis_defense_game.html**: game page and interface.
- **thesis_defense_game.js**: state, rendering, collision, scoring.
- **style.css**: shared blog design and game layout.

Game Loop

- Update player movement and target position.
- Move threats toward their target objects with tile-based route selection.
- Let gate archers shoot matching attackers using cooldowns and arrows.
- Check sword rescues, collisions, citations, and pass/fail pressure.
- Redraw the maze, player, archers, humanoid threats, arrows, particles, and HUD.

AI-ASSISTED DEVELOPMENT

LLMs helped turn ideas into engineering tasks

1. Planning

Converted the prompt into a static web-game architecture suitable for GitHub Pages.

2. Building

Implemented character selection, Canvas rendering, maze movement, pathfinding, archers, sword rescue, and score logic.

3. Debugging

Balanced difficulty, checked wall collision, win/fail rules, canvas dimensions, local links, and responsive layout.

4. Documentation

Prepared the report, reflection, and presentation material.

The main lesson: AI is most useful when the request is specific, testable, and connected to a clear software goal.

Stability checks before publishing

What I Checked

- JavaScript syntax check passed.
- Maze rows and columns match the canvas grid.
- All local page links resolve correctly.
- Desktop and mobile layouts render properly.
- Game shows explicit pass or failure messages.

Demo Flow

- Open Assignment 4 page.
- Click the playable game entry.
- Select a character and start the round.
- Show gate archers shooting matching attackers.
- Rescue a pinned archer with the player's sword.
- Defend Bugs, Deadlines, and Peer Reviewers to show weakening.
- Miss a threat to show growth and rejection pressure.

REFLECTION

A small but complete AI-assisted application

- This project made the difference between a fun prototype and a stable application clearer.
- A playable game needs state management, input control, rendering, collision detection, pathfinding, helper AI, scoring, clear failure rules, responsive design, and deployment.
- My final result meets the Option A requirements and is ready to be hosted from RenXuan's blog.

Conclusion: **Defend Your Thesis** shows how LLMs can support real software development when combined with testing and engineering judgment.